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10th International Mathematics Assessments for Schools (2020-2021)

Junior Division Round 2

Time: 120 minutes

Printed Name _____ *Code* _____ *Score* _____

Instructions:

- Do not open the contest booklet until you are told to do so.
- Be sure that your name and code are written on the space provided above.
- Round 2 of IMAS is composed of three parts; the total score is 100 marks.
- Questions 1 to 5 are given as a multiple-choice test. Each question has five possible options marked as A, B, C, D and E. Only one of these options is correct. After making your choice, fill in the appropriate letter in the space provided. Each correct answer is worth 4 marks. There is no penalty for an incorrect answer.
- Questions 6 to 13 are a short answer test. Only Arabic numerals are accepted; using other written text will not be honored or credited. Some questions have more than one answer, as such all answers are required to be written down in the space provided to obtain full marks. Each correct answer is worth 5 marks. There is no penalty for incorrect answers.
- Questions 14 and 15 require a detailed solution or process in which 20 marks are to be awarded to a completely written solution. Partial marks may be given to an incomplete presentation. There is no penalty for an incorrect answer.
- Use of electronic computing devices is not allowed.
- Only pencil, blue or black ball-pens may be used to write your solution or answer.
- Diagrams are not drawn to scale. They are intended as aids only.
- After the contest the invigilator will collect the contest paper.

The following area is to be filled in by the judges;
the contestants are not supposed to mark anything here.

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	Total Score	Signature
Score																	
Score																	

Junior Division Round 2

Questions 1 to 5, 4 marks each

1. If $P=\frac{2021}{2020}-\frac{2020}{2021}$ and $Q=\frac{1}{2020}\times\frac{1}{2021}$, then what is the value of $\frac{P}{Q}$?

（A）1　（B）2　（C）2020　（D）2021　（E）4041

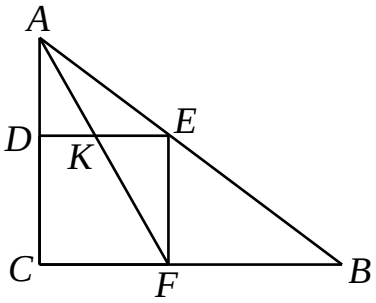
Answer：_____

2. How many non-negative integers x satisfy $|3x+2|\le 20$?

（A）6　（B）7　（C）13　（D）19　（E）20

Answer：_____

3. In the figure, ABC is a right triangle such that $\angle ACB=90°$, $AC=6$, $CB=8$ and $CDEF$ is a square. If AF meets DE at point K, then what is the value of $\frac{DK}{EK}$?



（A）$\frac{3}{4}$　（B）$\frac{3}{5}$　（C）$\frac{9}{16}$　（D）$\frac{\sqrt{3}}{2}$　（E）$\frac{7}{15}$

Answer：_____

4. Emily bought some number of fruit candies and milk candies for a total of \$100, where each fruit candy costs \$3 while each milk candy costs \$8. It is known that the number of fruit candies bought is less than the number of milk candies bought. How many candies did Emily bought in total?

（A）13　（B）15　（C）20　（D）25　（E）Cannot be determined

Answer：_____

5. Given that $x^2=x+1$, which of the following algebraic equations is equal to x^8?

（A）$4x+4$　（B）$8x+5$　（C）$13x+8$

（D）$16x+10$　（E）$21x+13$

Answer：_____

Questions 6 to 13, 5 marks each

6. Define $(a, b, c, d) = ab - cd$, where a, b, c and d are positive integers.

   Given that $\dfrac{(a, 3, 2, 1)}{(3, 2, 1, a)} = 5$, determine the value of a.

   Answer : _____

7. For any real number x, we define $[x]$ to be the greatest integer that does not exceed x and $\{x\}$ to be the fractional part of x such that $\{x\} = x - [x]$.

   If $2[x] + 3\{x\} = \dfrac{69}{5}$, then what is the value of x?

   Answer : _____

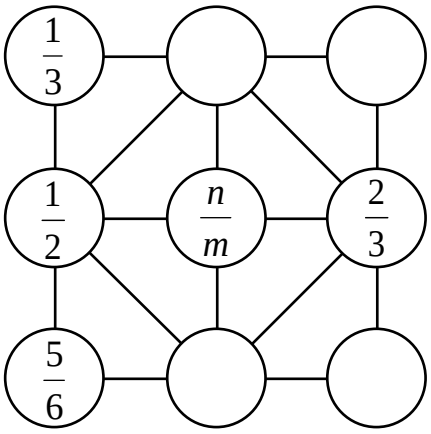
8. If $xyz > 0$ so that $x + y = \dfrac{xy}{2}$, $y + z = \dfrac{yz}{3}$ and $z + x = \dfrac{zx}{4}$, then what is the value of z?

   Answer : _____

9. Let a, b and c be positive integers such that $a > b > c$. It is known that b is a single digit number, c is a single digit prime number and $a^2 + b^2 + c^2 = 2021$. What is the value of $a + b + c$?

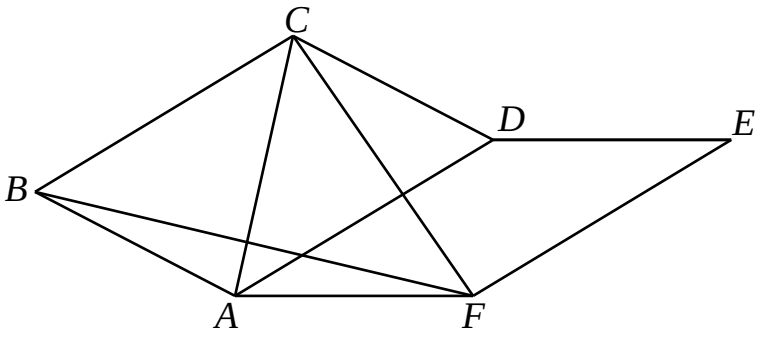
   Answer : _____

10. In the diagram, fill each of the "◯" with numbers so that the sum of the four vertices of all squares that can be found are equal. If the middle number is filled as $\dfrac{n}{m}$, where m and n are both positive numbers that are relatively prime. What is the value of $m + n$?



   Answer : _____

11. In the figure below, *ABCD* and *ADEF* are both parallelograms and the areas of triangles *ABF* and *ACF* are 20 cm^2 and 50 cm^2, respectively. What is the area, in cm^2, of parallelogram *ADEF*?



Answer： _____ cm^2

12. Let *a*, *b*, *c* and *d* be positive integers such that $a > b > c > d$ which satisfy the following system of equations:

$$\begin{cases} a+b+c+d=2020 \\ a^2-b^2+c^2-d^2=2020 \end{cases}$$

How many ordered quadruples (*a*, *b*, *c*, *d*) are there that satisfy the conditions?

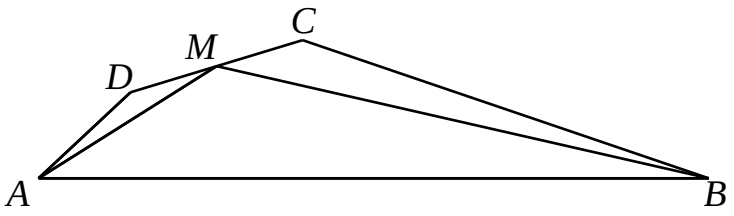
Answer： _____ ordered quadruples

13. Let $\overline{abcd}$ and $\overline{cdab}$ be two 4-digit numbers such that *a*, *b*, *c* and *d* are four different digits from 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. If *M* is the greatest common factor of both $\overline{abcd}$ and $\overline{cdab}$, then what is the largest possible value of *M*?

Answer： _____

Questions 14 to 15, 20 marks each
(Detailed solutions are needed for these two problems)

14. In the figure below, $ABCD$ is a convex quadrilateral such that M is the midpoint of CD and $\angle AMB = 135°$. Prove that $AD + BC + \frac{\sqrt{2}}{2} CD \geq AB$.



15. There exists a positive integer N such that $N+1$ is a multiple of 24. Prove that the sum of all the positive factors of N is also a multiple of 24.